

Intent Cognitive Synergy (ICS)

From the Internet of Everything to an Intelligent Digital Life Form



Research Achievements of the Network Intelligence Research Center (NIRC)
Prof. Jingyu Wang & Team | Beijing University of Posts and Telecommunications
Building a Soul for the Digital World.

The Vision: Lighting the Fire.

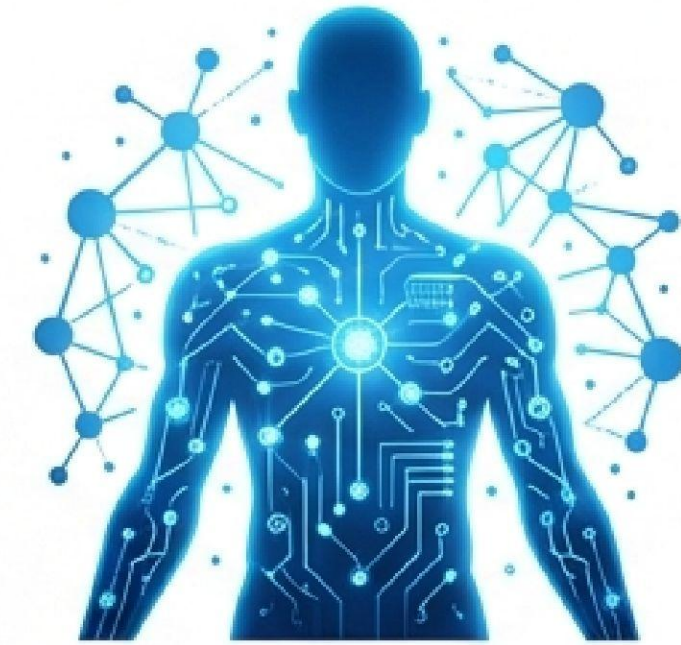
*"If the traditional internet is a **warehouse of parts**, we are **lighting the fire**... eventually endowing everything with a wise soul." — Prof. Jingyu Wang*



Cold Hardware
(Internet of Everything)



Intent Perception



Digital Life Form
(Intent Cognitive Synergy)

Identity

The IE Team
(Intelligence of
Everything).

Leadership

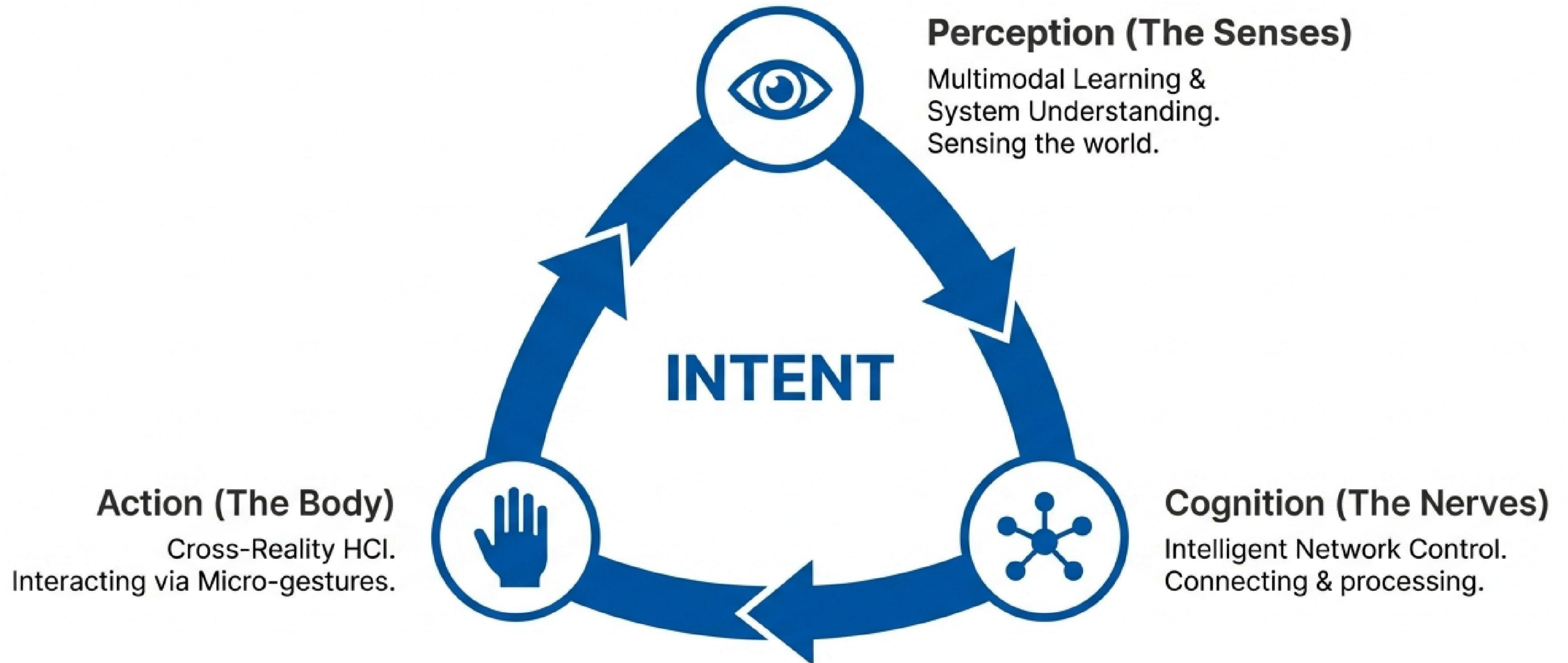
Prof. Jingyu Wang
(CIC Fellow, National
High-level Talent).

Impact

- 300+ Papers (NSDI, SIGCOMM, CVPR, AAAI)
- 15+ Provincial Applications
- National Tech Progress Award (2nd Class)

Anatomy of an Intelligent Digital Life Form

The Intent Cognitive Synergy (ICS) Loop



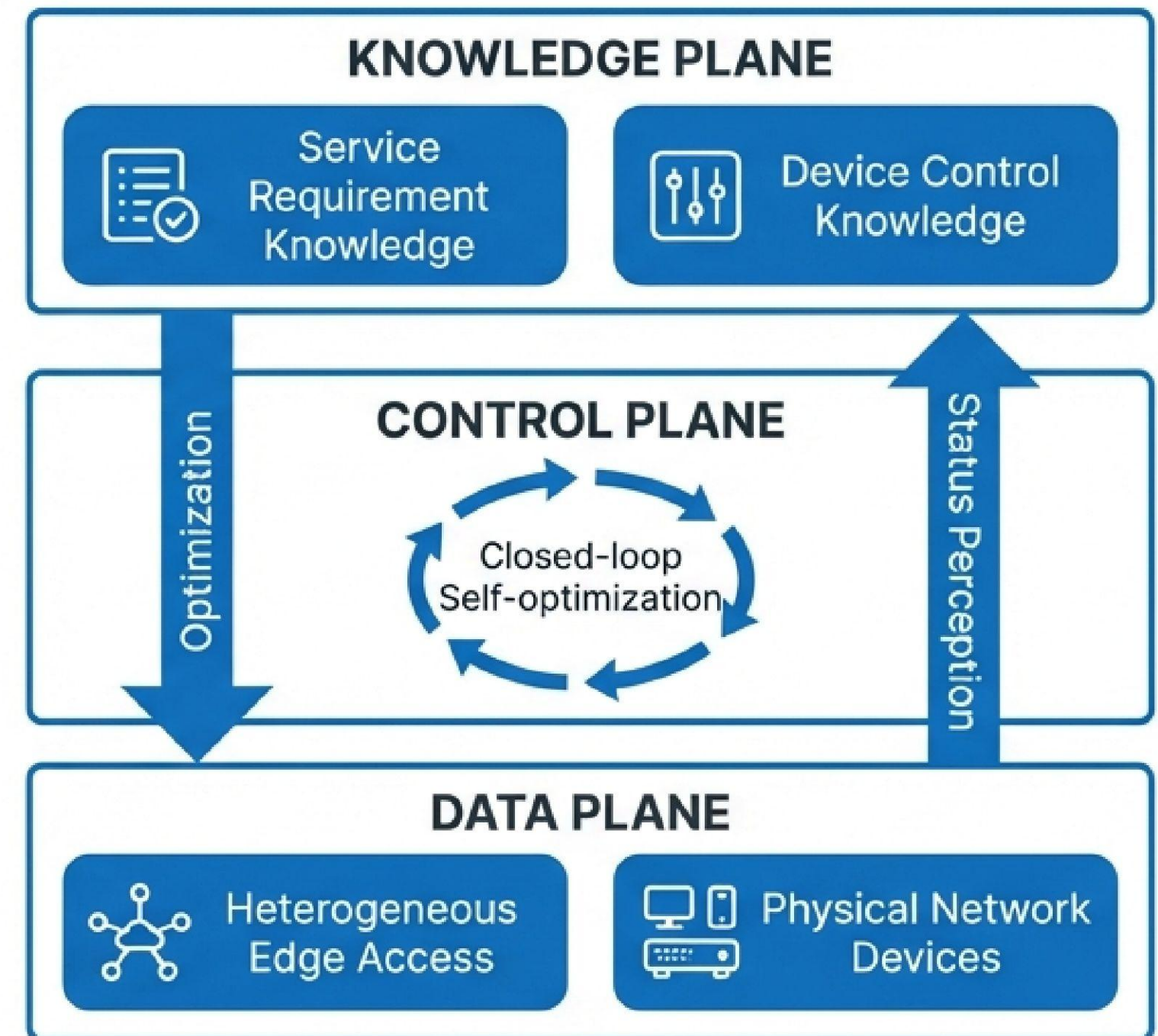
Pillar I: The Nervous System (Intelligent Network Control)

Knowledge-Defined Networking (KDN)

The Shift:

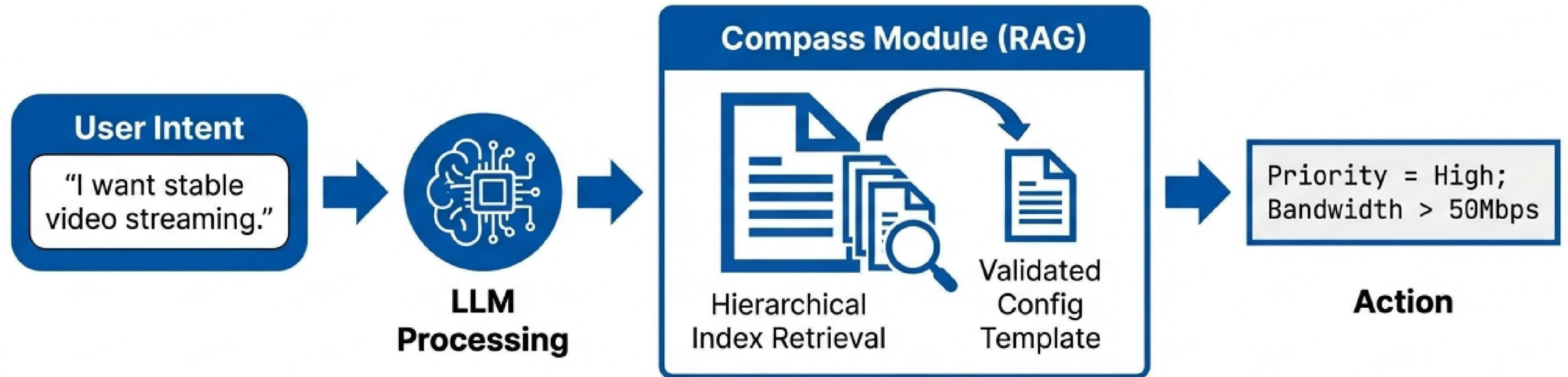
Traditional: Rigid, manual configuration.

NIRC Innovation: Intent-Driven Synergy. The network self-configures based on high-level goals.



Following the Compass: LLM-Empowered Intent Translation

Solving LLM Hallucination in Critical Infrastructure

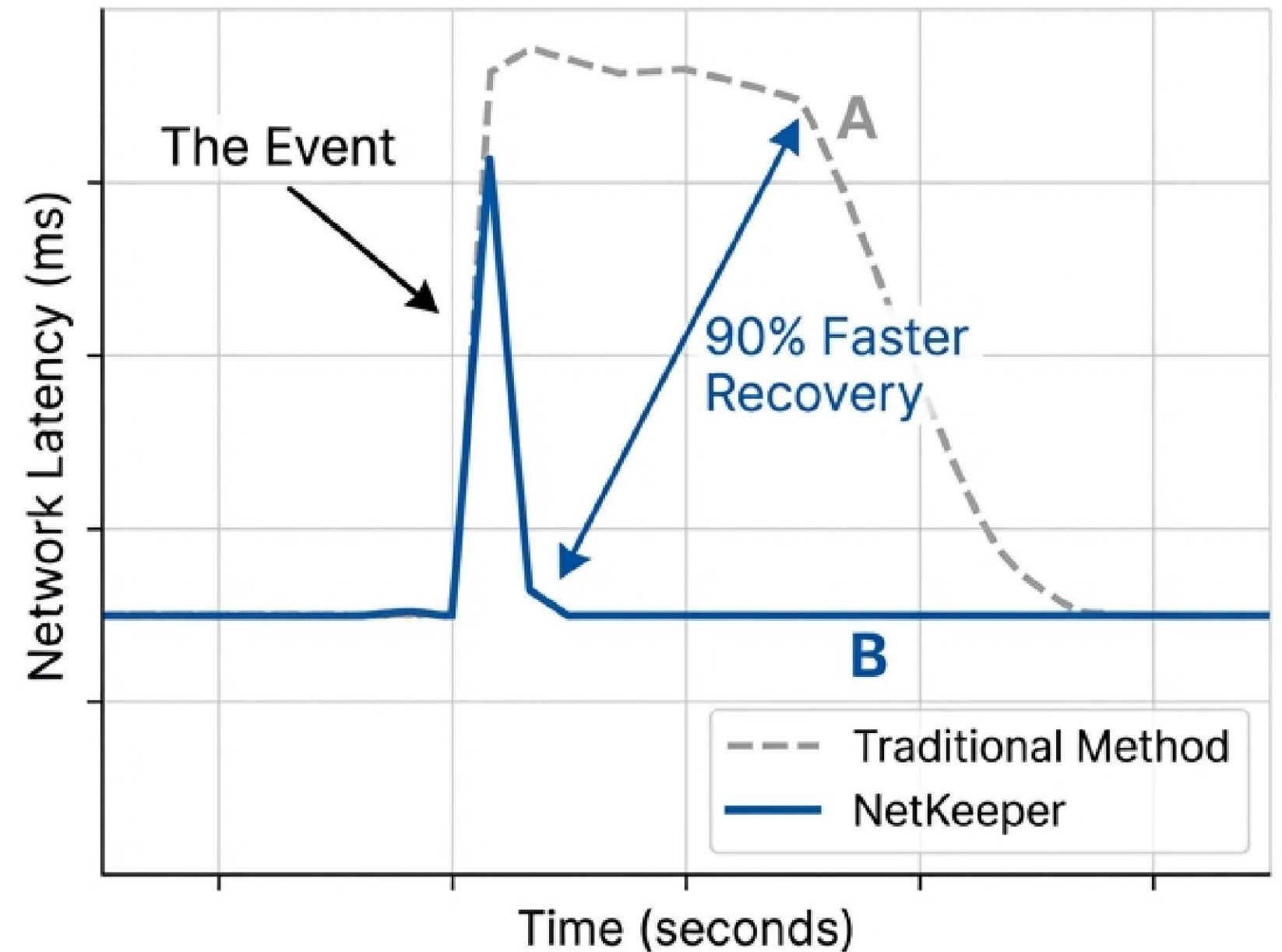


- **Problem:** LLMs hallucinate; networks require precision.
- **Solution:** Retrieval-Augmented Generation (RAG) + Manual Guidance. Compass retrieves accurate indexes to ground the AI.

NetKeeper: Autonomous Network Resilience

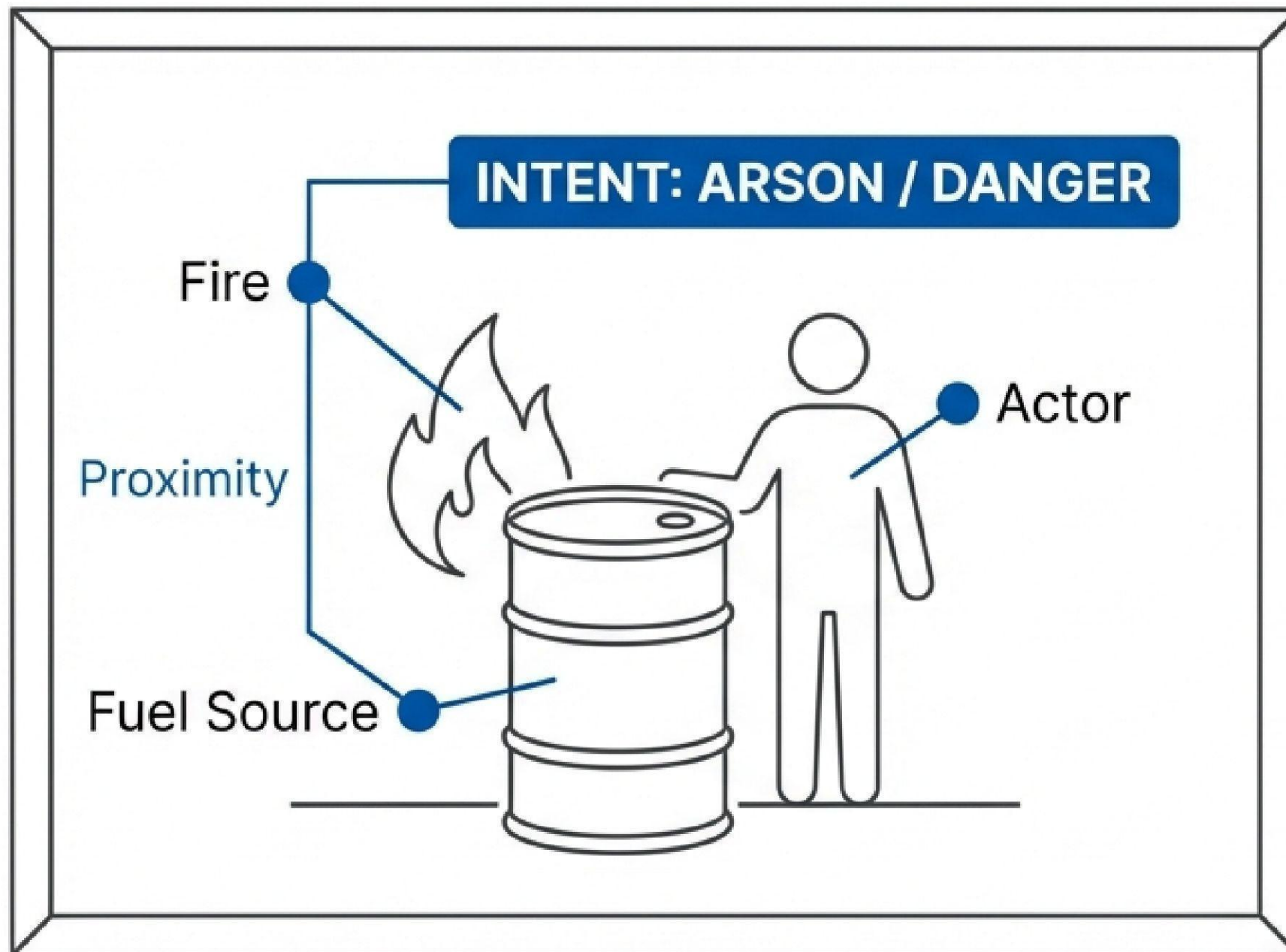
Self-Healing Configuration Updates

- **Context:** Digital life forms must survive traffic bursts and attacks.
- **Innovation:** Autonomous Configuration Update. Dynamic adjustment based on real-time traffic patterns.
- **Deployment:** Scale application in Massive IoT data handling.



Pillar II: The Senses (Video & Multimodal Perception)

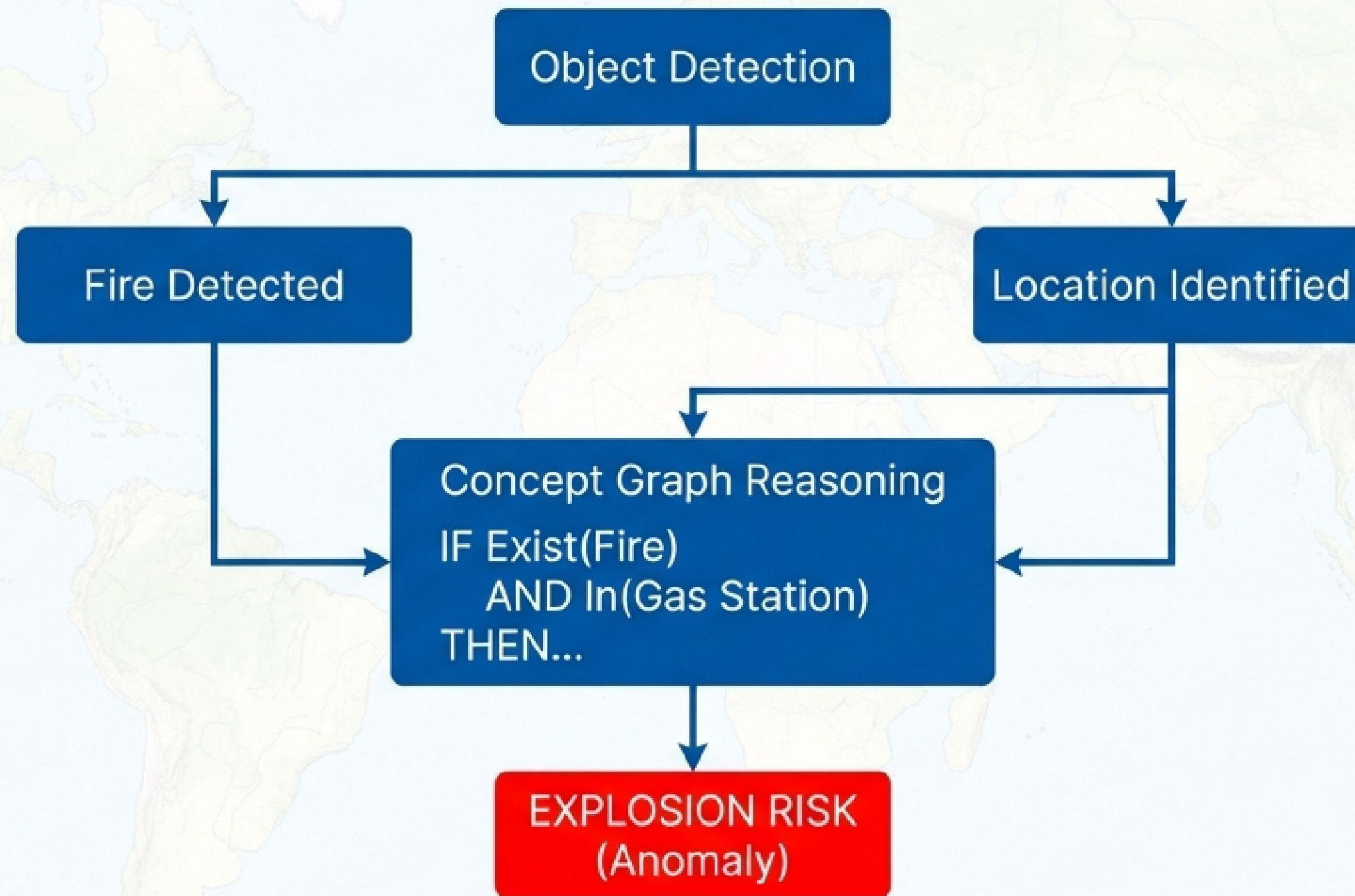
From Pixels to Semantics: Understanding Context



Standard cameras see movement.
ICS sees **intent**.

Merging **Explicit Knowledge (Logic)** with **Implicit Data (Pixels)** to distinguish harmless events **from critical anomalies**.

CG-VAD: Knowledge-Defined Anomaly Detection



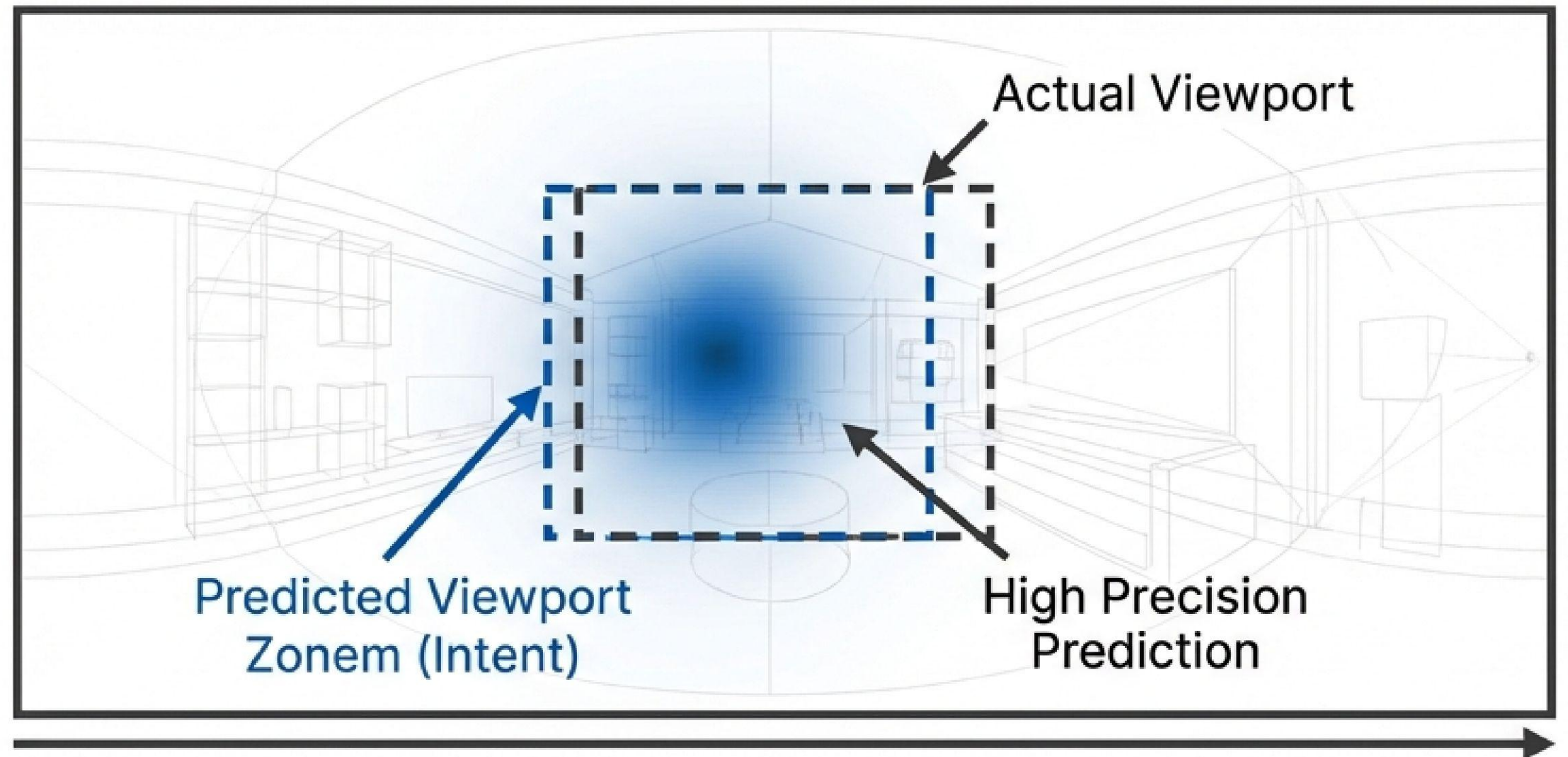
Innovation: Concept Graphs define logical relationships. The system reasons about the 'intent' of the scene (e.g., Fire + Gas Station = Danger) rather than just pixel changes.

Predicting the User's Gaze: PsyQoE & STAR-VP

Intent-Aware Video Streaming

Context: In 360° video, bandwidth is wasted on unseen areas.

Solution: STAR-VP predicts where the user *intends* to look using space-time fusion and psychological assessment.

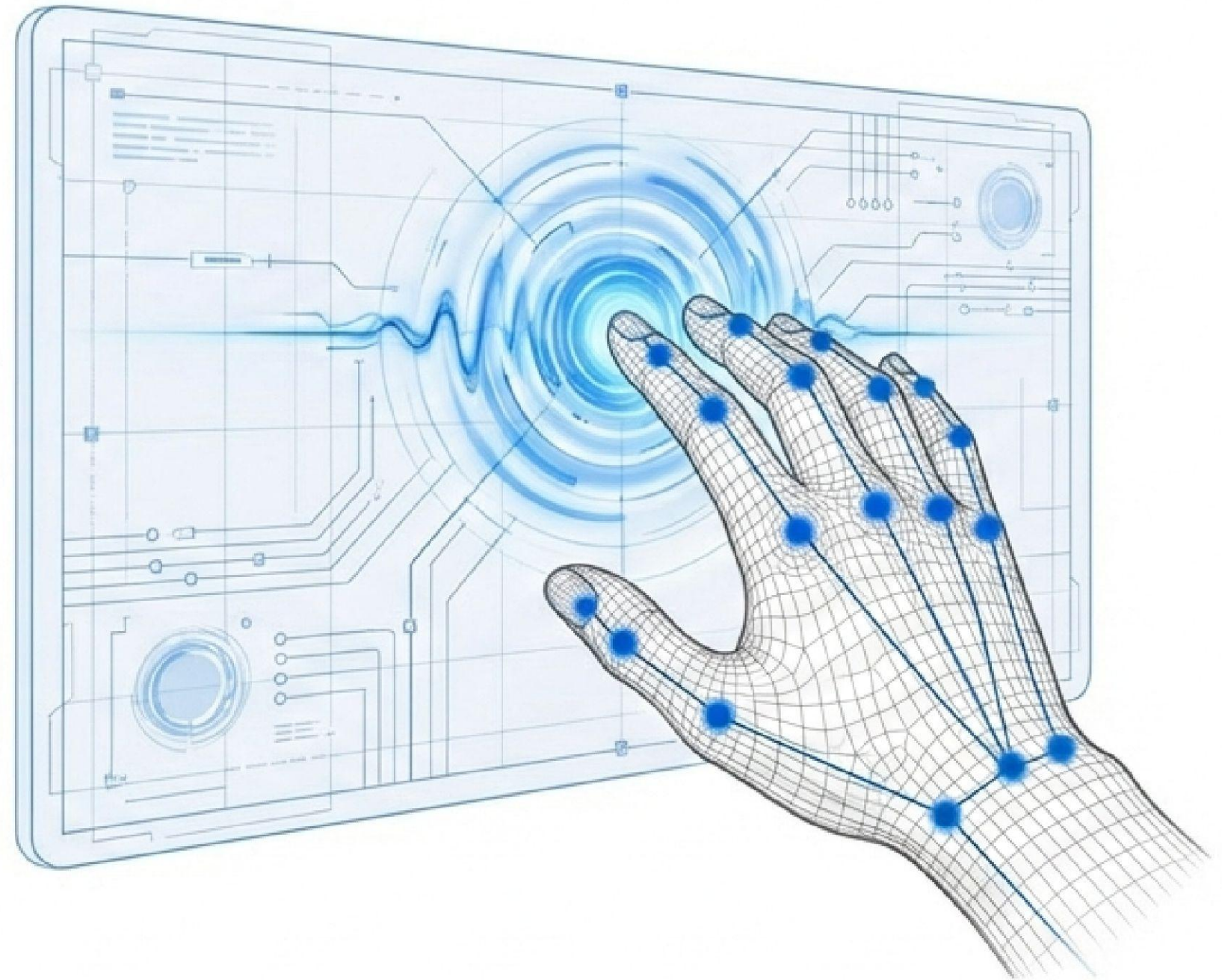


Pillar III: The Interface (Cross-Reality HCI)

The Digital Hand: Zero-Latency Interaction

Philosophy: “What I gesture is what I get.” Removing physical controllers.

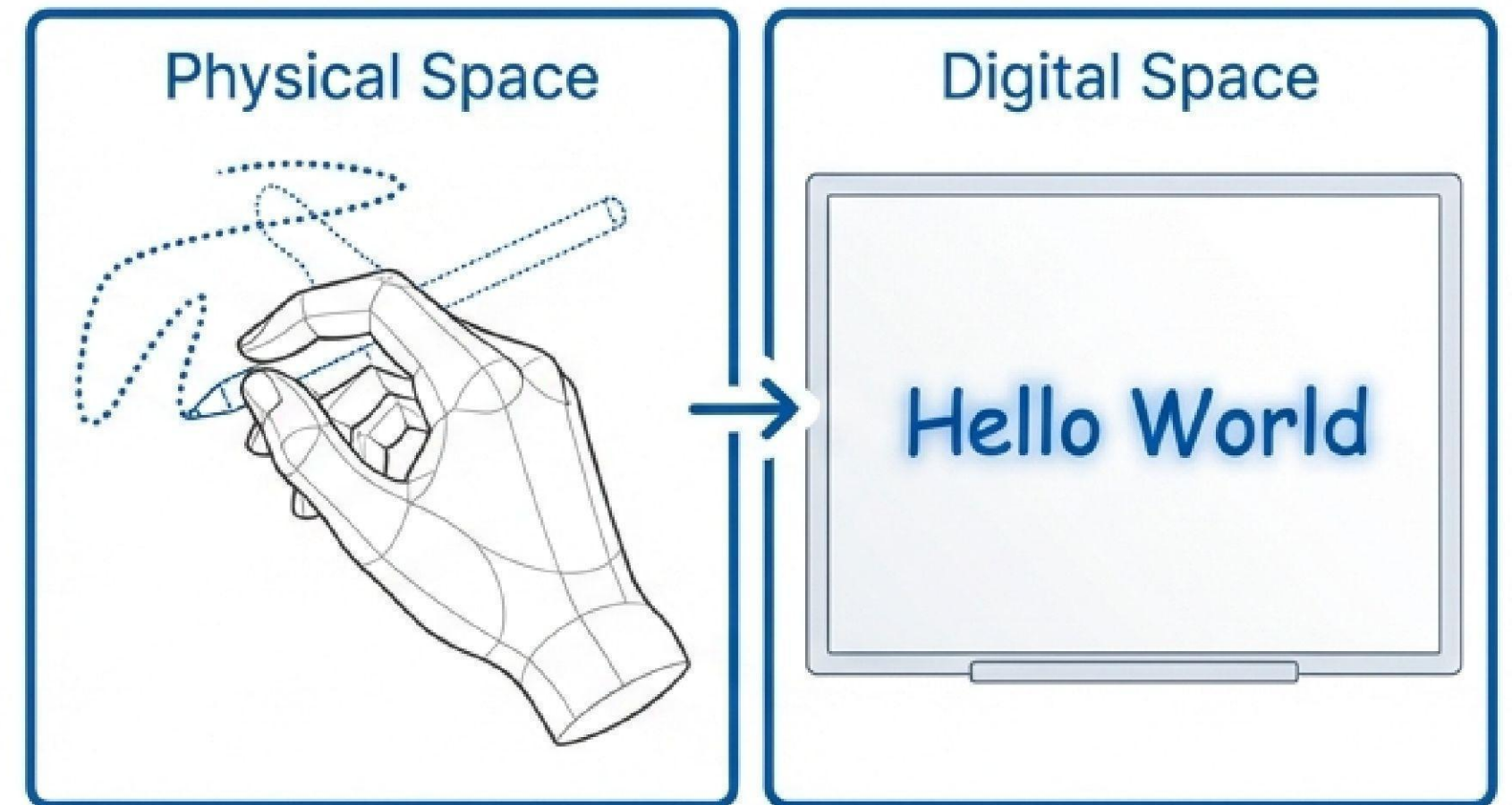
Key Tech: MegoTrack (ICCV 2025 Champion). Multi-view bare-hand tracking with pre-calibrated shape precision.



InkFlow & FreeWriting: Writing in Thin Air

Innovation: XR Virtual Whiteboard.
Supports natural bare-hand writing
without a physical surface.

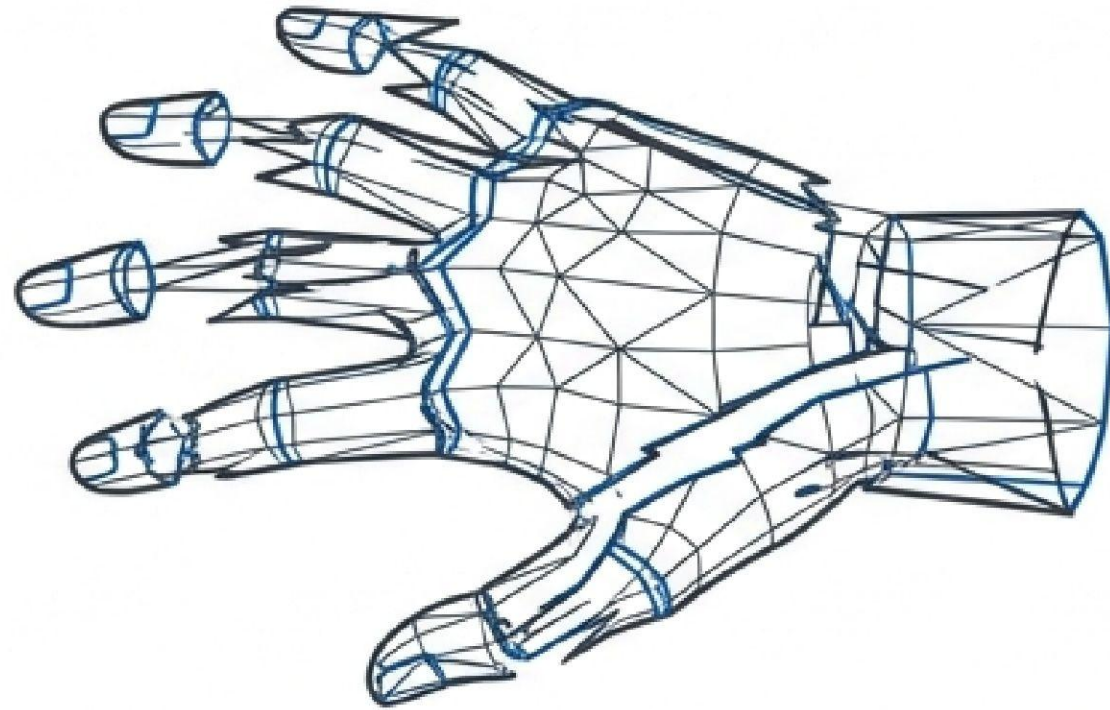
Intent Aspect: The system predicts
the 'intent' of the micro-gesture
to smooth out jitter and produce
clean, legible ink.



Generative Hand Avatars: Temporal & Content Co-Awareness

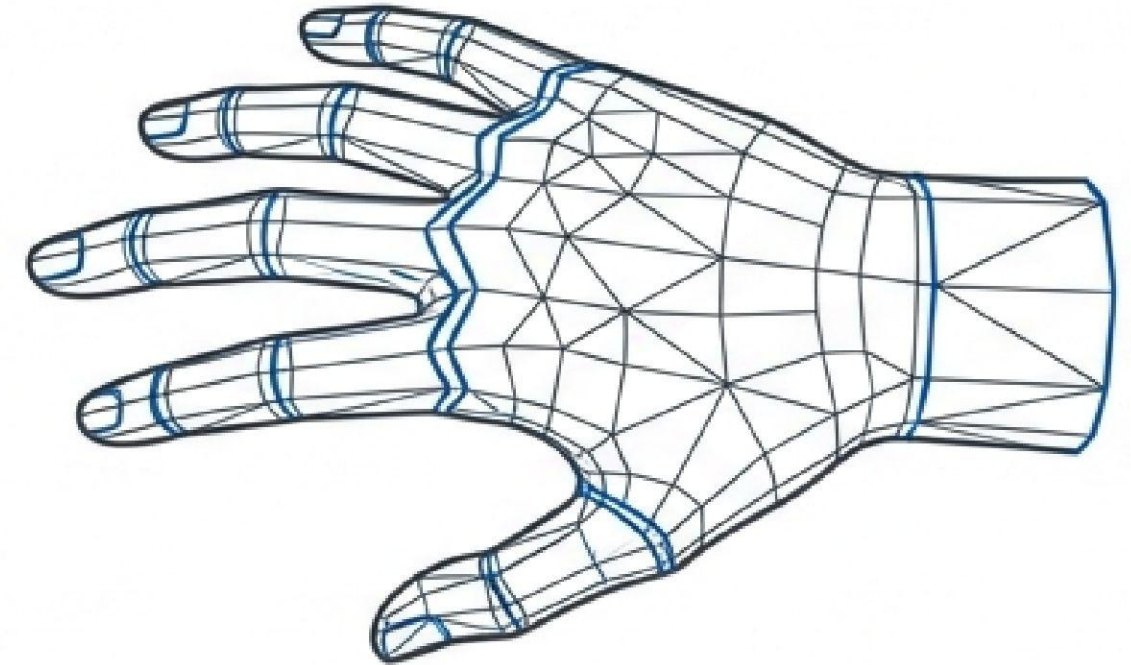
Solving the “Broken Mesh” Problem in Virtual Worlds

Standard Generation



Latent Diffusion
Co-Awareness

NIRC Method [CVPR 2026]



Problem: Virtual hands often glitch or clip during complex movements.

Solution: Latent Diffusion for Controllable Hand Generation ensures temporal consistency and content integrity.

Industrial Impact & Recognition

Awards



National Science and Technology Progress Award (2nd Class)



AAAI 2023 Distinguished Paper Award



IEEE Systems Journal Best Paper Award (2021)

Deployment

- **Scale:** Deployed in 15+ Provincial Units.
- **Impact:** Handling massive IoT data streams and security logs.

Partners

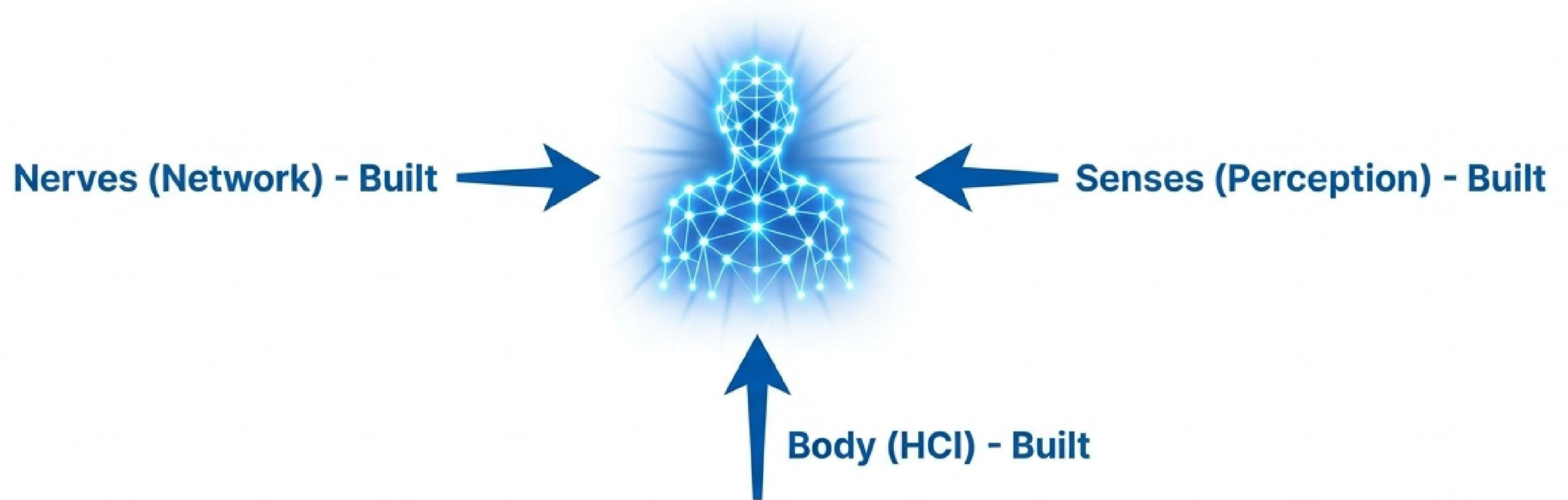


国家电网
STATE GRID



美团

The Future: Closing the Loop



The Final Frontier: Theoretical Computer Science.
Provable. Explainable. Generalizable.

Intent-Driven Synergy: Building a Soul for the Digital World.

NIRC Team | <https://nirc.top/> | GitHub: OpenNetAI